SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 4

Plan, Observe, Replan



The Planning Pattern: the model lays out an ordered investigation plan, then revises it when a new observation breaks an assumption.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

A plan that can change its mind

1

What planning adds

An ordered path for the whole question, not one step.

2

Before AI

Examiners scope and sequence before they collect.

3

The problem

A first plausible timeline, treated as final.

4

The discipline

Revise the path when evidence breaks an assumption.

5

The gap

A plan built on partial evidence can mislead.

6

Planning

Plan, observe, replan, then conclude.

7

The case

A missing phone, and a late network record.

8

Your lab

Plan from part of the evidence, then replan.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

WHERE WE ARE

ReAct took one step. Planning sees the path.

Lab 3 · ReAct

The model chose the next single step after seeing the last result. Good for one check at a time.

But it never looked at the whole question up front.

TODAY

Lab 4 · Planning

Now the model lays out an ordered plan for the whole investigation, then revises the remaining steps as evidence arrives.

It thinks about the path, not just the next footstep.

THE IDEA, WITHOUT THE FORENSICS

A plan you revise when a step fails

THE FIRST PLAN

1. estimate guests
2. choose a date
3. reserve a room
4. order food

THE ROOM IS TAKEN -> REPLAN

1. check availability
2. pick another date or room
3. update the guest count
4. order food

REPLANNING IS THE POINT

A planner does not order food before it knows how many people can come. When a step fails, it reorders what is left. In this lab, the “room is taken” moment is a network record that changes everything.

THE EXAMINER'S HABIT

Scope and sequence before you collect

Define the scope

Fix the incident window before gathering anything.

Order the subgoals

Decide which records to pull first, and why.

Follow dependencies

Later steps use what earlier steps established.

Revise when needed

A surprising record reshapes the remaining plan.

A PLAN IS A HYPOTHESIS ABOUT ORDER

A seasoned examiner writes the investigation sequence before touching the data, precisely so a later surprise forces a conscious revision rather than a quiet assumption. Planning teaches a model to work the same way.

THE PROBLEM

The first plausible timeline, treated as final

THE PLAN THAT NEVER CHANGES

“There was a call at 21:08 and a WhatsApp image at 21:13, so both messages were completed during the missing period.”

WHY A FIXED PLAN FAILS HERE

It ignores what it has not seen

The first plan never asked about connectivity at all.

A new record changes the meaning

The phone was offline from 21:10, around the attachment.

Activity is not delivery

An image record inside the window is not a sent message.

THE DISCIPLINE

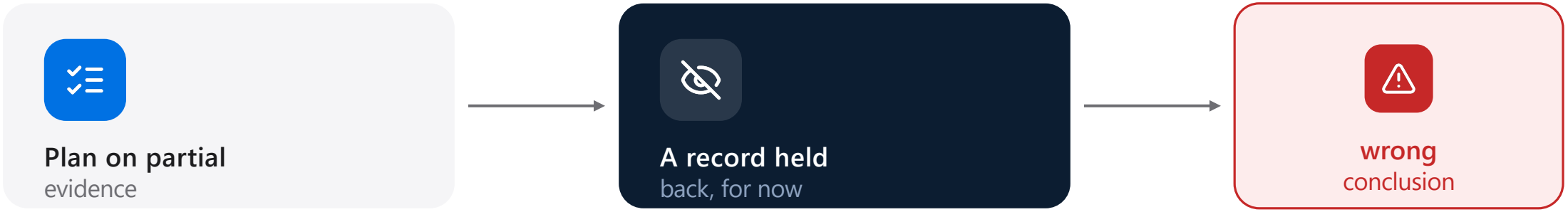
Revise the path, do not restart it

1	**Expect to be wrong**	Write the first plan with named triggers for what would change it.
2	**Watch each observation**	Compare every new record against the plan's assumptions.
3	**Reorder, keep what holds**	On a conflict, revise the remaining steps; do not scrap good ones.
4	**Bound the conclusion**	Let the final claim shrink to what the revised plan supports.

Replanning is not failure. It is the plan doing its job.

THE GAP

A plan is only as good as its evidence

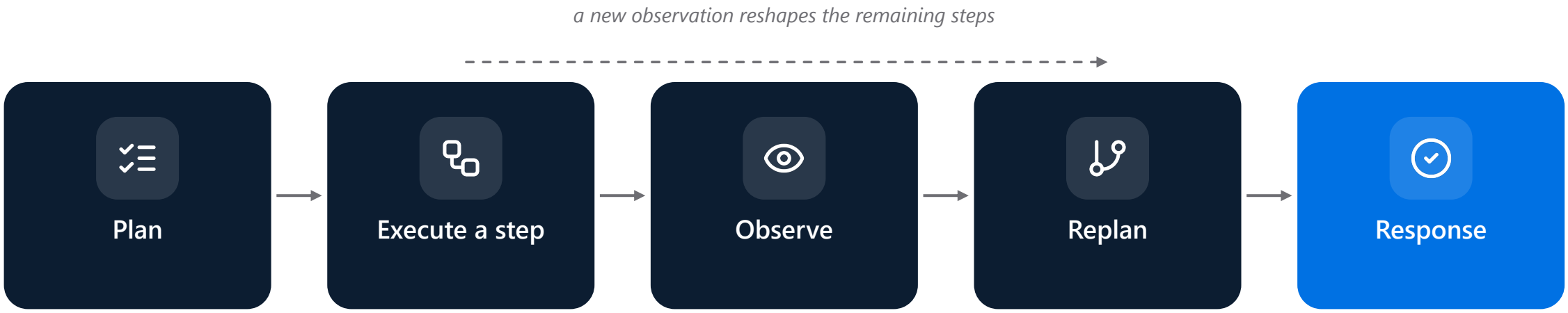


THE WHOLE CASE TURNS ON ONE WITHHELD RECORD

In this lab the network log is deliberately held back until after the first plan. That mirrors real work, where artifacts surface late. A planner that cannot revise will carry its early, uninformed assumptions straight into the conclusion. ReAct reacts to the next step; planning has to re-examine the whole path.

THE ANSWER

The Planning Pattern



PLAN FIRST, THEN ADAPT

The plan sets the order up front. Each observation can reshape the steps that remain.

WHEN DOES IT STOP?

When the remaining plan is sufficient and no observation conflicts, the model writes the final conclusion.

BUILD THE ALARM IN FROM THE START

Name your replanning triggers

A good plan says, up front, what would make it wrong. Those conditions are replanning triggers.

A missing dependency

A later step needs a fact no earlier step produced.

A contradiction

A new record disagrees with something the plan assumed.

A timing conflict

An event lands outside the window the plan was built on.

IN THIS CASE

The trigger that fires is a timing-and-contradiction one: the network log shows the phone was offline exactly when the WhatsApp attachment was recorded.

THE INVESTIGATION

Case PL-2026-011

THE DEVICE

PHONE

Google Pixel 8, Android 14

ACQUIRED

Cellebrite UFED, full file system

THE REPORT

phone missing for ~30 minutes

THE TASK

reconstruct the timeline

THE QUESTION

Which communication events happened inside the 30-minute window, and what can we say about delivery?

A call

An outgoing call at 21:08, inside the window.

A WhatsApp record

A chat and an image attachment near 21:13, also inside.

A late record

The phone was offline 21:10 to 21:26, around the attachment.

THE SEQUENCE, WITH THE TWIST

Six events, one offline gap

20:55:03	device unlocked
21:08:14	outgoing call, 42 seconds
21:10:00	mobile data goes offline
21:12:07	WhatsApp chat opened
21:13:12	image-attachment record, then deleted
21:25:31	device locked
21:27:04	mobile data restored

The offline window (21:10 to 21:26) overlaps the attachment.
That overlap is what turns a “delivery” into an “attempt.”

WATCH THE PLAN CHANGE ITS MIND

Before and after the network record

Initial plan (partial evidence)

- Set the window from unlock records.
- Place the outgoing call at 21:08.
- Add the WhatsApp activity at 21:12-21:13.
- **Conclude: both communications completed.**

Revised plan (network added)

- Keep the window, call, and WhatsApp steps.
- Insert: check connectivity around 21:13.
- Find the device offline 21:10-21:26.
- **Conclude: in-window activity, delivery unconfirmed.**

PRACTICAL

Your turn, in the browser

1	**Open in Colab**	Click the badge in the notebook. Nothing to install.
2	**Paste a free key**	Get a free Groq key, paste it when asked. It is hidden.
3	**Plan on partial evidence**	Part A: build the first plan before the network record.
4	**Replan on the new record**	Add the network log, then revise and conclude.
5	**Compare PlanningAgent**	Part B: run the packaged agent and judge its revision.

SectraTEK

The best plan is one you revise.

Next lab: Multiagent, where several specialist agents split the work, then reconcile their findings into one judgment.

